

Monte Carlo simulation of π^- interactions on a polyethylene target and optimization of the target thickness

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Abstract

A Monte Carlo simulation of a high-energy π^- beam interacting with a polyethylene target is presented. The study focuses on the optimization of the target thickness in order to maximize the reconstruction performance for neutral mesons η and π^0 decaying into two photons.

1 Introduction

Monte Carlo simulations are a useful tool in experimental high-energy physics, where detector design and analysis strategies must be optimized before data taking. By generating synthetic events according to known physical laws and detector responses, Monte Carlo methods allow to study acceptance effects, reconstruction efficiencies and statistical performance in a controlled environment.

In this work, a Monte Carlo simulation of a π^- beam with an energy of 100 GeV impinging on a polyethylene target is developed. The signal processes $\pi^- p \rightarrow \pi^0 n$ and $\pi^- p \rightarrow \eta n$, followed by the decay $\pi^0, \eta \rightarrow \gamma\gamma$, are considered. The main goal of the study is the optimization of the target thickness L , in order to maximize the statistical significance of the reconstructed signal while preserving a good invariant-mass resolution.

2 Physics Process and Experimental Setup

The incident beam is assumed to be monoenergetic, composed of π^- with a fixed energy of 100 GeV, collimated along a fixed direction. Effects related to beam divergence or energy spread are neglected.

The target material is polyethylene (CH_2), which is approximated as a source of free protons at rest for hadronic interactions. The target is assumed as a homogeneous medium with a variable thickness L , expressed in cm.

For the purpose of the optimization, the signal is defined in an operational sense. Only processes leading to the production of a neutral meson (π^0 or η) decaying into two photons are considered as signal contributions. All other interactions are treated as background, independently of their underlying dynamics.

The detector is assumed to be sensitive exclusively to electromagnetic final states. As a consequence, the only experimentally accessible objects in the simulation are the two photons originating from the neutral-meson decay. No information on the recoiling neutron or on the primary interaction vertex is available for the reconstruction. The optimization procedure is therefore entirely based on the reconstruction of the two-photon final state.

The observable used to identify the signal is the invariant mass of the photon pair, $m_{\gamma\gamma}$. Neutral mesons are reconstructed as peaks in the $m_{\gamma\gamma}$ distribution over a smooth background. The performance of the measurement, in terms of statistical significance and mass resolution, depends on the target thickness through a balance between interaction probability, photon absorption and detector resolution effects.

The simulation assumes a single relevant interaction per incident pion and neglects secondary hadronic interactions inside the target, as well as pile-up effects. These approximations are sufficient for isolating the impact of the target thickness on the reconstruction of neutral mesons from their two-photon decay.

2.1 Target and detector geometry

The target is modeled as a cylindrical volume of radius of 10 cm, with its axis aligned along the beam direction. The interaction length L corresponds to the cylinder depth along the beam axis.

The detector is represented by an idealized electromagnetic calorimeter aligned along the beam direction located at 10 m from the target. It is modeled as a circular surface with radius of 100 cm, centered on the beam axis. Only photons intersecting the detector surface are considered detectable. The geometrical acceptance of the detector therefore plays a crucial role in the reconstruction efficiency and is explicitly taken into account in the simulation.

3 Monte Carlo Simulation Framework

The simulation is organized as a modular Monte Carlo chain, in which each step corresponds to a well-defined physical or detector process. The main stages include the sampling of the interaction point inside the target, the generation of the scattering kinematics, the decay of neutral mesons, and the simulation of the detector response to the final-state photons.

3.1 Interaction Depth Sampling

The first step of the Monte Carlo simulation consists in sampling the depth at which the incident π^- undergoes its first hadronic interaction inside the target. As the pion traverses the material, the probability of interaction per unit length is governed by the total hadronic cross section on the target nuclei.

Under the assumption of a homogeneous target, the survival probability of a pion propagating a distance z without interacting follows an exponential law. The probability density for the first interaction to occur at depth z is therefore given by

$$P(z) = \frac{1}{\lambda} \exp\left(-\frac{z}{\lambda}\right), \quad (3.1.1)$$

where λ is the interaction length, defined as

$$\lambda = \frac{1}{n \sigma_{\text{tot}}}, \quad (3.1.2)$$

with n the number density of scattering centers and σ_{tot} the total interaction cross section.

Considering the polyethylene molecule (CH_2), and assuming that only protons (hydrogen nuclei) contribute to the scattering process, n is given by:

$$n_p = 2 \cdot \frac{\rho}{M} N_A \approx \frac{0.93}{14} \cdot 6.02 \cdot 10^{23} \text{cm}^{-3} \approx 8.00 \cdot 10^{22} \text{cm}^{-3} \quad (3.1.3)$$

where ρ is the material density, N_A is Avogadro's number, and M is the molar mass of polyethylene. The factor of 2 accounts for the two hydrogen atoms (i.e. two protons) present in each molecule, which are the only scattering centers considered in this study.

The total π^-p cross section at a beam energy of 100 GeV is set at the nominal value of:

$$\sigma_{\text{tot}}(\pi^-p, 100\text{GeV}) \approx 25\text{mb} \quad (3.1.4)$$

This data is taken from the Particle Data Group compilation [1], which is based on earlier experimental measurements such as those reported in Ref. [2].

In conclusion the interaction length estimated is:

$$\lambda \approx 499.95 \text{ cm}$$

Given this probability density, the probability for a pion to undergo at least one interaction within a target of thickness L is obtained by integrating the distribution over the target length,

$$P_{\text{int}}(L) = \int_0^L \frac{1}{\lambda} e^{-x/\lambda} dx = 1 - e^{-L/\lambda}. \quad (3.1.5)$$

When $L \ll \lambda$ the expression can be approximated as:

$$P_{\text{int}}(L) \approx \frac{L}{\lambda} \quad (3.1.6)$$

In the simulation, the interaction depth z is sampled according to the exponential distribution introduced above using the inverse transform method.

Since the cumulative distribution function $F(z) = P(Z \leq z)$ is uniformly distributed in the interval $[0, 1]$, one can write

$$u = F(z) = 1 - e^{-z/\lambda}, \quad u \sim \mathcal{U}(0, 1).$$

Inverting the relation yields

$$z = -\lambda \ln(1 - u).$$

Because $1 - u$ is also uniformly distributed in $[0, 1]$, this expression can be equivalently written as

$$z = -\lambda \ln u.$$

The variable z is therefore used as the interaction position along the target.

This represents a valid approximation, as the decay lengths of the produced π^0 or η mesons are negligible compared to the characteristic interaction length $\lambda \sim \text{cm}$. Even in the extreme case where the full beam energy is transferred to the produced meson, one finds

$$L_{\pi^0} \simeq \frac{E}{m} c\tau \approx 19 \mu\text{m}, \quad L_{\eta} \simeq \frac{E}{m} c\tau \approx 27 \text{ nm},$$

with $c\tau$ from PDG^{3.1.1}. These lengths are several orders of magnitude smaller than the relevant length scale of the problem.

Only interactions occurring within the physical target length, $0 < z < L$, are retained. If the sampled depth exceeds the target thickness, the pion is assumed to exit the target without interacting.

3.1.1 Validation of the interaction-depth sampling

The behavior of the interaction-depth sampling has been validated through a set of dedicated control plots, shown in Appendix A.

These plots are intended to verify the internal consistency of the sampling procedure, illustrate the dependence of the interaction probability on the target thickness, and assess the impact of approximations introduced in simplified models.

First, the number of simulated first interactions as a function of the target thickness L is compared to the analytically expected behavior of eq. 3.1.5 showing very good agreement within statistical uncertainties, as it can be seen in fig. A.0.1. The same comparison is repeated using the linearized approximation of eq. 3.1.6 in fig. A.0.2, confirming the expected validity range of this approximation at small $L \ll \lambda$.

In addition, the distribution of the sampled interaction depth z for a fixed target thickness is compared to the corresponding truncated exponential probability density. In this case, small deviations between the Monte Carlo histogram and the analytical curve are observed and are expected, since the latter is compared to a binned estimator affected by finite statistics and binning effects, particularly in the tail of the distribution, as shown in fig. A.0.3.

Since the interaction depth is generated by direct inversion of the cumulative distribution function, the resulting distributions are analytically guaranteed to reproduce the target probability density.

The control plots therefore serve mainly as consistency checks and as a visual illustration of the expected physical behaviour.

3.2 Channel probabilities

From experimental measurements in Ref. [2], the following nominal cross sections @100 GeV are reported:

$$\sigma_{tot}(\pi^- p \rightarrow \pi^0 n) = 3.20 \mu\text{b}; \quad \sigma_{2\gamma}(\pi^- p \rightarrow \eta n) = 0.33 \mu\text{b}; \quad (3.2.1)$$

Once an interaction has occurred inside the target, the final state of the event is selected probabilistically according to the relative production cross sections of the available channels. In particular, the $\pi^- p \rightarrow \pi^0 n$ and $\pi^- p \rightarrow \eta n$ reactions are generated with probabilities

$$p_{\pi^0} = \frac{\sigma_{\pi^0}}{\sigma_{tot}} \approx 0.013\%, \quad p_{\eta} = \frac{\sigma_{\eta}}{\sigma_{tot}} \approx 0.001\% \quad (3.2.2)$$

where σ_{π^0} and σ_{η} denote the corresponding channel cross sections of eq. 3.2.1, and σ_{tot} is the total $\pi^- p$ interaction cross section at the beam energy of eq. 3.1.4. All remaining interaction channels, collectively referred to as background, are generated with probability

$$p_{\text{bkg}} = 1 - p_{\pi^0} - p_{\eta} \approx 99.9\%$$

^{3.1.1}For η the PDG actually gives $\Gamma_{tot} = 1.31 \text{ keV}$, then one can find $\tau = \frac{\hbar}{\Gamma}$

The channel selection is implemented by sampling a uniform random number and assigning the event to one of the available final states according to these probabilities.

Since the production cross sections for the π^0 and η channels are much smaller than the total π^-p cross section, only a small fraction of the interactions lead to signal events. As a consequence, the expected number of π^0 and η events is much smaller than the total number of interactions and the total number of interactions inside the target is dominated by the background contribution.

The absolute signal yields are proportional to the total number of interactions in the target and therefore increase with the target thickness L until a saturation due to the asymptotic behavior of the interaction probability of eq. 3.1.5. This is illustrated in Fig. 3.2.1, where the expected counts for the π^0 and η channels are shown as a function of L .

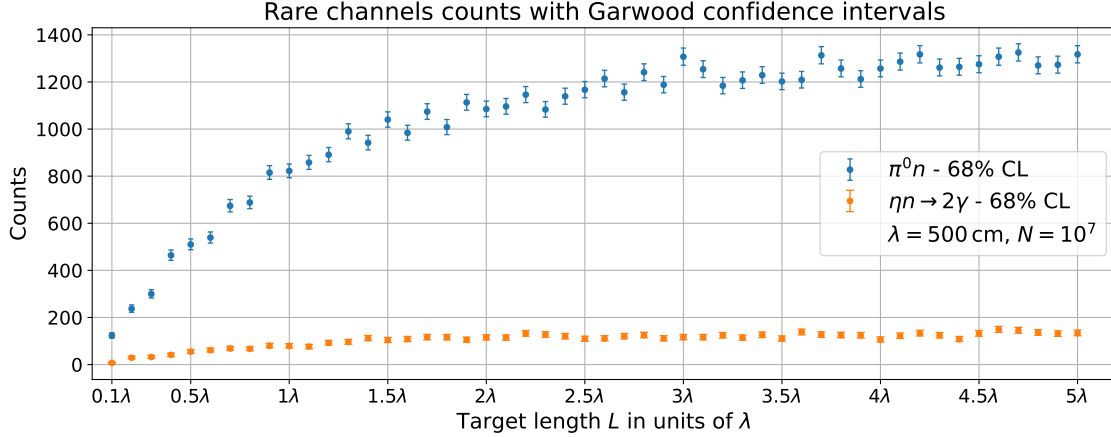


Figure 3.2.1: Expected event counts for the rare $\pi^-p \rightarrow \pi^0 n$ and $\pi^-p \rightarrow \eta n$ channels as a function of the target thickness L . Error bars represent 68% confidence intervals computed using the Garwood prescription for Poisson-distributed counts.

In the low-statistics regime relevant for rare channels, that are Poisson distributed event counts, it is more appropriate to characterize statistical uncertainties through interval estimates rather than by associating errors from maximum-likelihood estimation (the commonly quoted $\sqrt{n}$, which becomes inaccurate physically for small n).

The uncertainties are then obtained through Poisson likelihood: confidence intervals are computed using the Garwood construction, which defines asymmetric intervals by inverting the cumulative Poisson distribution, ensuring proper coverage also in the low-count regime^{3.2.1}.

3.3 Meson angular distribution from differential cross-section

In the process $\pi^-p \rightarrow Xn$, the information on the angular distribution in the center-of-mass frame is encoded in the differential cross section

$$\frac{d\sigma}{dt},$$

which is experimentally measured and provided in tabulated form in Ref. [3] for π^0 and in Ref. [4] for η as a function of the variable

$$\tau \equiv -t \geq 0.$$

The tabulated values of $d\sigma/dt$ carry physical dimensions ($\mu\text{b}/\text{GeV}^2$) and therefore represent a cross-section density.

For each bin i of the tabulated distribution, the following quantities are known:

- the bin-averaged value

$$y_i \equiv \left(\frac{d\sigma}{dt} \right)_i,$$

- the bin boundaries in τ

$$\tau_i^{\text{low}} \leq \tau \leq \tau_i^{\text{high}}, \quad \Delta\tau_i = \tau_i^{\text{high}} - \tau_i^{\text{low}}.$$

^{3.2.1}For an explicit and statistically formally in-depth analysis see the Appendix B

The cross section associated with bin i is approximated by

$$\Delta\sigma_i \simeq y_i \Delta\tau_i,$$

corresponding to the bin area in the $(\tau, d\sigma/dt)$ plane. This approximation assumes that the differential cross section is constant within each bin.

The probability for an event to fall into bin i is defined as the fraction of the total cross section carried by that bin,

$$P_i = \frac{\Delta\sigma_i}{\sum_j \Delta\sigma_j} = \frac{y_i \Delta\tau_i}{\sum_j y_j \Delta\tau_j}.$$

This normalization transforms the experimentally measured differential cross section into a discrete probability distribution over the bins, satisfying

$$\sum_i P_i = 1.$$

Therefore the resulting probability density function for τ can be built, and it turns out to be piecewise constant,

$$p(\tau) = \begin{cases} \frac{P_i}{\Delta\tau_i} = \frac{y_i}{\sum_j y_j \Delta\tau_j} & \tau \in [\tau_i^{\text{low}}, \tau_i^{\text{high}}], \\ 0 & \text{otherwise} \end{cases}$$

To enable Monte Carlo sampling, a discrete cumulative distribution function is defined from the bin probabilities,

$$\text{CDF}_k = \sum_{i=1}^k P_i, \quad \text{CDF}_N = 1,$$

where N denotes the total number of bins.

This cumulative distribution is defined on the discrete bin index rather than on the continuous variable τ . The value CDF_k represents the probability that an event belongs to a bin with index less than or equal to k , i.e. the probability accumulated after fully integrating bin k . As a consequence, the discrete CDF increases in steps at bin boundaries and remains constant within each bin. It is therefore used exclusively for the selection of the bin index in the sampling procedure.

3.3.1 Sampling algorithm

Finally the sampling of $\tau = -t$ is performed in two stages:

1. A uniform random number is drawn,

$$u \sim \mathcal{U}(0, 1).$$

2. The first bin index i satisfying

$$\text{CDF}_i \geq u$$

is selected. This step assigns bin i with probability P_i .

3. Once the bin is selected, the continuous variable τ is sampled uniformly within the bin,

$$\tau \sim \mathcal{U}(\tau_i^{\text{low}}, \tau_i^{\text{high}}),$$

consistently with the assumption of a constant differential cross section inside the bin.

4. Finally, the momentum transfer is obtained as

$$t = -\tau.$$

The correctness of this implemented sampling procedure has been verified through dedicated control plots, which compare the piecewise-constant probability density function reconstructed from the tabulated $d\sigma/dt$ data with a large Monte Carlo sample of generated values of t as shown in Fig. 3.3.1a and 3.3.1b.

These plots also provide a direct visualization of the effective angular distribution used in the simulation.

In particular, the strong concentration of events at small values of $|t|$ clearly illustrates the forward nature of the meson production at high energy, which has important consequences for detector acceptance and event reconstruction.

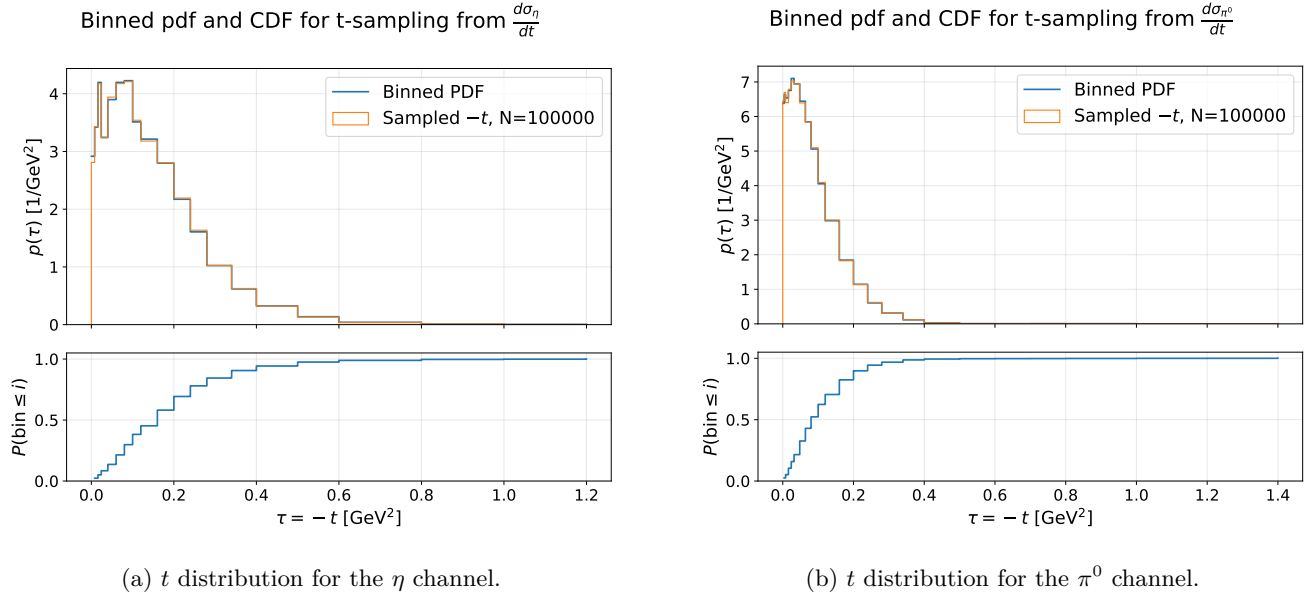


Figure 3.3.1: Piecewise-constant probability density functions in $\tau = -t$ reconstructed from the tabulated $d\sigma/dt$ data (solid lines), together with Monte Carlo samples generated via inversion of the corresponding discrete cumulative distribution functions.

3.4 Event Kinematics

This section describes the kinematic modeling of the process

$$\pi^- + p \rightarrow X + n, \quad X = \pi^0, \eta,$$

followed by the decay

$$X \rightarrow \gamma\gamma,$$

The full event kinematics is completely determined by energy-momentum conservation, the sampled momentum transfer t , and the isotropic decay of the meson in its rest frame.

All the used angular variables followed these conventions:

- The polar angle θ is measured with respect to the z axis, which is defined as the direction of the incoming π^- beam.
- The azimuthal angle ϕ is measured in the transverse (x, y) plane around the beam axis, starting from the x axis and increasing counterclockwise.
- The pseudorapidity is defined as

$$\eta = -\ln \tan \frac{\theta}{2},$$

and therefore depends only on the polar angle θ with respect to the beam direction. Large values of η correspond to particles emitted at small polar angles (forward region), while particles emitted at large angles populate smaller values of η .

3.4.1 Two-body scattering $\pi^- p \rightarrow Xn$

The incoming proton is assumed to be at rest in the laboratory frame,

$$P_p = (m_p, 0, 0, 0), \quad P_\pi = (E_\pi, 0, 0, p_\pi),$$

with

$$\beta_{\text{CM}} = \frac{|\vec{p}_{\text{tot}}|}{E_{\text{tot}}} = \frac{p_\pi}{m_p + E_\pi}.$$

along the beam direction or z axis.

The Mandelstam invariant s is given by

$$s = (P_p + P_\pi)^2 = m_p^2 + m_\pi^2 + 2m_p E_\pi.$$

In the center-of-mass frame, the total momentum vanishes and the final-state particles are produced back-to-back with equal momentum magnitude p^* . The latter is fixed by relativistic kinematics,

$$p^* = \frac{\sqrt{\lambda(s, m_X^2, m_n^2)}}{2\sqrt{s}}, \quad \lambda(a, b, c) = a^2 + b^2 + c^2 - 2ab - 2ac - 2bc.$$

After sampling the momentum transfer t ^{3.4.1} from the differential cross section $d\sigma/dt$, the polar scattering angle in the center-of-mass frame is obtained as

$$\cos \theta^* = \frac{t - m_\pi^2 - m_X^2 + 2E_\pi^* E_X^*}{2p_\pi^* p^*},$$

while the azimuthal angle is sampled uniformly,

$$\phi^* \sim \mathcal{U}(0, 2\pi).$$

The three-momentum of the produced meson in the center-of-mass frame is therefore

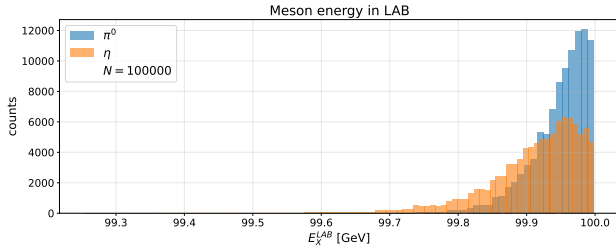
$$\vec{p}_X^* = p^* (\sin \theta^* \cos \phi^*, \sin \theta^* \sin \phi^*, \cos \theta^*).$$

A Lorentz boost with velocity $\vec{\beta}_{\text{CM}}$ is then applied to obtain the meson four-momentum in the laboratory frame,

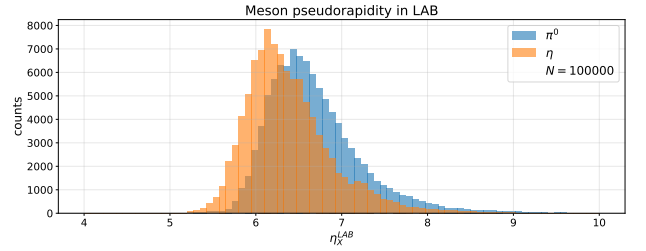
$$p_X^{\text{LAB}} = \Lambda(\vec{\beta}_{\text{CM}}) p_X^*.$$

The resulting laboratory-frame energy and pseudorapidity distributions of the produced mesons are shown in Fig. 3.4.1. Because the beam energy is much larger than the meson masses and the scattering is strongly forward, the meson carries almost the full beam energy. The energy distribution is therefore sharply peaked close to $E_X^{\text{LAB}} \simeq E_\pi$, with a slightly broader tail for the heavier η meson.

At fixed center-of-mass energy, the lighter π^0 is on average more strongly boosted along the beam direction than the η . As a consequence, the π^0 pseudorapidity distribution is shifted toward larger values compared to that of the η .



(a) Laboratory-frame energy of the produced mesons.



(b) Laboratory-frame pseudorapidity of the produced mesons.

Figure 3.4.1: Energy (left) and pseudorapidity (right) distributions of the produced π^0 and η mesons in the laboratory frame.

3.4.2 Decay $X \rightarrow \gamma\gamma$

The decay $X \rightarrow \gamma\gamma$ is generated in the rest frame of the meson. Since the decay is isotropic and fully constrained by energy-momentum conservation, the two photons are emitted back-to-back with fixed energy

$$E_\gamma^{(X)} = \frac{m_X}{2}.$$

^{3.4.1}The Mandelstam variable t is defined as the squared four-momentum transfer between the incoming pion and the outgoing meson,

$$t \equiv (p_\pi - p_X)^2 = m_\pi^2 + m_X^2 - 2(E_\pi E_X - \vec{p}_\pi \cdot \vec{p}_X),$$

and is negative in the physical region considered here.

The photon directions are sampled as

$$\cos \theta_\gamma \sim \mathcal{U}(-1, 1), \quad \phi_\gamma \sim \mathcal{U}(0, 2\pi),$$

defining the unit vector $\hat{n}$ as

$$\hat{n} = (\sin \theta_\gamma \cos \phi_\gamma, \sin \theta_\gamma \sin \phi_\gamma, \cos \theta_\gamma).$$

The photon four-momenta in the meson rest frame are then

$$k_{1,2}^{(X)} = \left(\frac{m_X}{2}, \pm \frac{m_X}{2} \hat{n} \right).$$

To obtain the photon four-momentum in the laboratory frame, a Lorentz boost with

$$\vec{\beta}_X = \frac{\vec{p}_X^{\text{LAB}}}{E_X^{\text{LAB}}}$$

is applied. The laboratory-frame photon energy can be written explicitly as

$$E_\gamma^{\text{LAB}} = \gamma_X E_\gamma^{(X)} \left(1 + \vec{\beta}_X \cdot \hat{n} \right), \quad \gamma_X = \frac{E_X^{\text{LAB}}}{m_X}.$$

The resulting photon energy and pseudorapidity distributions are shown in Fig. 3.4.2, which shows the two photons have identical distributions when averaged over many events.

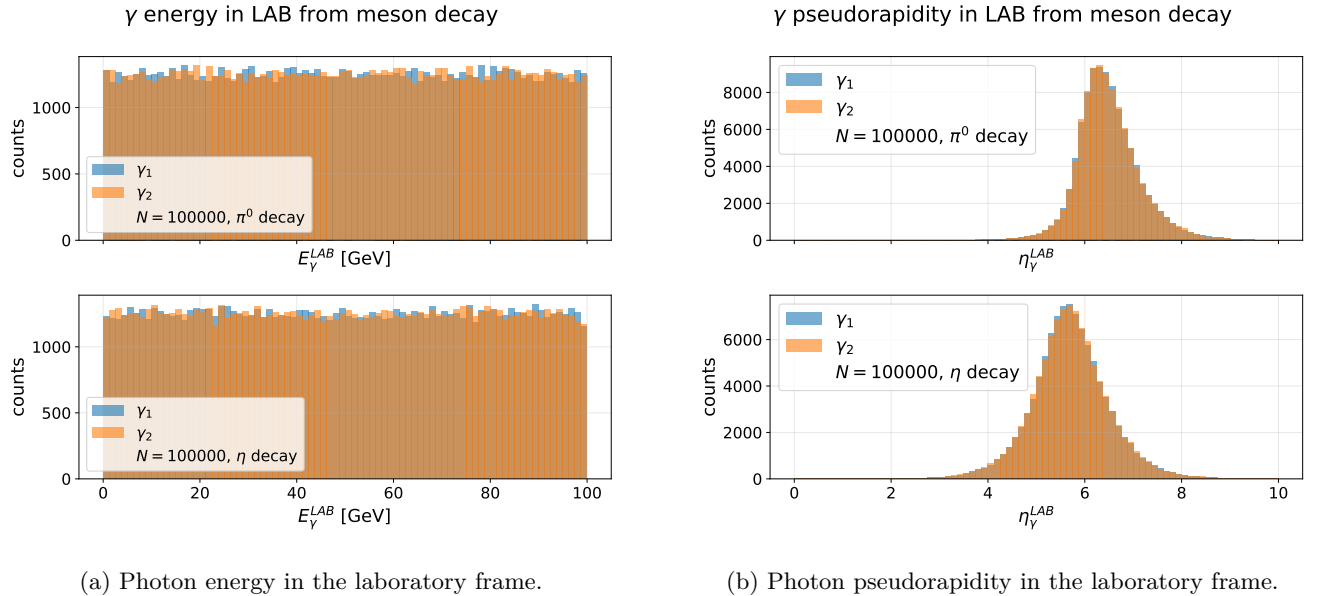


Figure 3.4.2: Energy (left) and pseudorapidity (right) distributions of photons from $X \rightarrow \gamma\gamma$ decays. Photons from π^0 decays are on average more forward than those from η decays, reflecting the larger boost of the lighter meson.

In detector coordinates, the angular separation between the two photons is quantified by

$$\Delta R = \sqrt{(\Delta\eta)^2 + (\Delta\phi)^2}, \quad (3.4.1)$$

where $\Delta\phi$ is computed taking into account the periodicity of the azimuthal angle. The corresponding three-dimensional opening angle α is defined by

$$\cos \alpha = \hat{n}_1 \cdot \hat{n}_2.$$

Figure 3.4.3 shows the ΔR distribution for photons from π^0 and η decays. Because the π^0 is significantly more boosted, its decay photons are more collimated and populate smaller ΔR values. In contrast, photons from η decays are less collimated, resulting in a broader separation distribution. This difference has direct implications for photon cluster separation and reconstruction in the detector.

3.5 Detector Response

The detector response is simulated by modeling the propagation, survival and reconstruction of the two photons produced in the decay $X \rightarrow \gamma\gamma$.

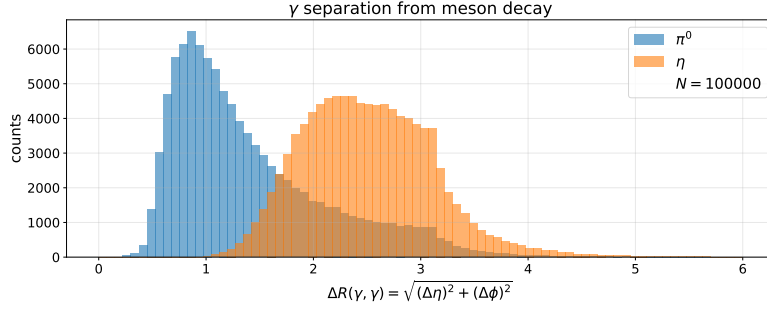


Figure 3.4.3: Distribution of the angular separation ΔR between the two photons from $X \rightarrow \gamma\gamma$ decays. Photons from π^0 decays are more collimated than those from η decays due to the larger boost of the parent meson.

3.5.1 Geometrical acceptance.

Each photon is propagated from the production point along a straight-line trajectory defined by its laboratory-frame direction $\hat{n}_\gamma$. The detector is geometrically modeled as already described in Section 2.1.

A photon is geometrically accepted if its trajectory intersects the detector surface. Denoting by (x_γ, y_γ) the transverse coordinates of the photon impact point on the detector plane, the geometrical acceptance condition can be written as

$$A_{\text{geom}}(\gamma) = \Theta(R_{\text{cal}}^2 - x_\gamma^2 - y_\gamma^2),$$

where Θ is the Heaviside step function. Only photons satisfying $A_{\text{geom}}(\gamma) = 1$ are considered detectable.

3.5.2 Photon escape probability.

Before reaching the calorimeter, photons must traverse a path length ℓ inside the target material. The photon path length is determined geometrically by the minimum between the forward exit through the downstream face of the target and a possible lateral exit through the cylindrical surface.

The probability for a photon to exit the target without undergoing an interaction is modeled as an exponential survival probability,

$$P_{\text{esc}}(\ell) = \exp\left(-\frac{\ell}{\lambda_\gamma}\right),$$

where λ_γ is the effective photon interaction length in the material.

As shown by the photon energy distributions in the laboratory frame in Fig. 3.4.2a, the photons produced in π^0 and η decays typically carry energies of several tens of GeV. In this energy regime, photon interactions with matter are dominated by electron-positron pair production, while Compton scattering and the photoelectric effect are negligible. The relevant physical scale governing photon absorption is therefore the radiation length X_0 .

Using the high-energy approximation for pair production, the photon interaction length is taken as

$$\lambda_\gamma = \frac{9}{7} X_0,$$

with $X_0 = 50.31\text{cm}$ for polyethylene.

On this basis, the target thickness is scanned in the range $L \simeq 2\text{--}82\text{ cm}$, corresponding to a fraction of the radiation length for which photon absorption effects are significant but not dominant.

The escape probability is sampled independently for the two photons, and events in which at least one photon fails to escape the target are rejected.

3.5.3 Energy smearing.

Photons that escape the target and satisfy the geometrical acceptance are reconstructed with finite energy resolution. The measured photon energy is obtained by applying a Gaussian smearing to the true energy,

$$E_\gamma^{\text{meas}} \sim \mathcal{N}(E_\gamma^{\text{true}}, \sigma_E^2).$$

The relative energy resolution is parameterized as

$$\frac{\sigma_E}{E_\gamma^{\text{true}}} = \sqrt{\left(\frac{a}{\sqrt{E_\gamma^{\text{true}}}}\right)^2 + b_{\text{eff}}^2},$$

where a represents the stochastic calorimetric term and b_{eff} the effective constant term.

To account for the degradation induced by photon propagation inside the target, the constant term is taken to depend on the photon path length ℓ ,

$$b_{\text{eff}} = \sqrt{b^2 + \left(k_E \frac{\ell}{\lambda_\gamma}\right)^2},$$

where b is the intrinsic constant term and k_E controls the strength of the material-dependent smearing. This parameterization introduces an explicit dependence of the energy resolution on the amount of material crossed by the photon.

3.5.4 Position smearing.

An analogous Gaussian smearing is applied to the photon impact position on the calorimeter plane,

$$\vec{x}_\gamma^{\text{meas}} \sim \mathcal{N}(\vec{x}_\gamma^{\text{true}}, \sigma_x^2 \mathbf{I}).$$

The position resolution is modeled as

$$\sigma_x = \sigma_{x,0} \left(1 + k_x \frac{\ell}{\lambda_\gamma}\right),$$

where $\sigma_{x,0}$ is the intrinsic detector resolution and the second term accounts for the additional smearing induced by the photon propagation inside the target material.

As a result of these two smearing, even photons that escape the target without interacting exhibit a progressively degraded reconstruction quality as the target thickness increases, which directly impacts the invariant-mass resolution of the reconstructed neutral mesons.

3.5.5 Reconstruction-level cuts.

Finally, reconstruction-level selection criteria are applied. Each photon is required to satisfy a minimum measured energy,

$$E_\gamma^{\text{meas}} > E_{\text{thr}},$$

and the two photons must be sufficiently separated in the detector. The separation is quantified in (η, ϕ) space as ΔR defined in eq. 3.4.1 and events are accepted only if

$$\Delta R > \Delta R_{\text{min}}.$$

4 Event Reconstruction

The event reconstruction aims at extracting physically meaningful observables from the reconstructed photon pairs and at quantifying the performance of the detector-target system as a function of the target thickness L .

4.1 Invariant-mass reconstruction

Events passing the detector response are characterized by two photons reconstructed following all the various steps of the Monte Carlo simulation described in Section 3, obtaining the measured energies $E_1^{\text{meas}}, E_2^{\text{meas}}$ and directions $(\eta_1^{\text{meas}}, \phi_1^{\text{meas}})$ and $(\eta_2^{\text{meas}}, \phi_2^{\text{meas}})$. From these quantities, the four-momenta of the photons are reconstructed and the invariant mass of the pair is computed as

$$m_{\gamma\gamma}^2 = 2E_1^{\text{meas}} E_2^{\text{meas}} (1 - \cos \alpha), \quad (4.1.1)$$

where α is the opening angle between the two reconstructed photon directions.

This observable represents the only experimentally accessible quantity used to identify neutral mesons decaying into two photons.

4.2 Signal and background composition

Signal events originate from the decays $\pi^0, \eta \rightarrow \gamma\gamma$ and produce narrow resonant peaks in the $m_{\gamma\gamma}$ spectrum. The peak widths are entirely driven by detector effects, such as energy and position resolution, photon acceptance, and cluster reconstruction.

The background is modeled using a simplified toy generator that produces pairs of uncorrelated photons. Each background photon is generated independently, both in energy and direction. The photon energy is sampled from a log-uniform distribution in the range

$$E_\gamma \in [E_{\min}, E_{\max}] = [0.1, 50] \text{ GeV},$$

meaning that the logarithm of the energy is uniformly distributed,

$$\log E_\gamma \sim U(\log E_{\min}, \log E_{\max}).$$

Under this assumption, the corresponding probability density function in energy is

$$p(E_\gamma) = \frac{1}{E_\gamma \ln(E_{\max}/E_{\min})}, \quad E_{\min} \leq E_\gamma \leq E_{\max}, \quad (4.2.1)$$

i.e. a scale-free spectrum proportional to $1/E_\gamma$. Such a behavior is physically motivated as a minimal and conservative model for non-resonant photon backgrounds, which are typically dominated by soft emissions and do not exhibit a characteristic energy scale.

The sampling of the photon energy is performed using the inversion method. If $u \sim U(0, 1)$ is a uniform random number, the corresponding photon energy is obtained as

$$E_\gamma = E_{\min} \left(\frac{E_{\max}}{E_{\min}} \right)^u, \quad (4.2.2)$$

which ensures an exact sampling of the log-uniform distribution. For a more statistically formal explanation on the derivation of eq. 4.2.1 and eq. 4.2.2 look at Appendix C.

Photon directions are generated assuming a uniform azimuthal angle and a forward pseudorapidity,

$$\phi \sim U(-\pi, \pi), \quad \eta \sim U(2, 6),$$

reflecting the forward-peaked kinematics typical of high-energy hadronic interactions.

4.3 Mass-resolution extraction

The intrinsic invariant-mass resolution is extracted by fitting the signal-only $m_{\gamma\gamma}$ distributions with a Gaussian model,

$$S(m_{\gamma\gamma}) = A \exp \left[-\frac{(m_{\gamma\gamma} - \mu)^2}{2\sigma_m^2} \right].$$

The fit is performed iteratively. Starting from initial guesses for μ and σ_m , the fit range is restricted to $\mu \pm k\sigma_m$, and the procedure is repeated multiple times until stable values of the parameters are obtained. This iterative approach suppresses the impact of non-Gaussian tails and residual background contributions, yielding a robust estimate of the core mass resolution $\sigma_m(L)$.

4.4 Oversampling and physical normalization

As shown in Fig. 3.2.1, even for a very large incident beam sample ($\mathcal{O}(10^7)$ pions), the expected number of produced π^0 and η mesons is extremely small, at the level of a few tens of events or less over the full target-length range, due to their relative small cross sections.

In this regime, a fully inclusive event generation would lead to invariant-mass spectra dominated by empty bins, making the reconstruction of a meaningful $m_{\gamma\gamma}$ spectrum and the extraction of mass resolution and significance impractical, unless prohibitively large Monte Carlo samples were generated. For this reason, oversampling is not employed only to reduce statistical fluctuations, but also to construct spectra with sufficient population for shape studies, fitting procedures, and optimization scans.

Operationally, oversampling is implemented by decoupling the generation of signal and background events from

their physical production probabilities. For each target thickness L , dedicated Monte Carlo samples are generated for the $\pi^0 \rightarrow \gamma\gamma$, $\eta \rightarrow \gamma\gamma$, and background channels, with fixed and sufficiently large numbers of generated events. In this step, the channel cross sections and relative production probabilities are effectively set aside, and the simulation is conditioned on the occurrence of a given final state. In this framework, the number of reconstructed events in the oversampled spectra is governed exclusively by photon escape probability and detector reconstruction inefficiencies.

To recover physically meaningful yields, all quantities entering the analysis are rescaled after reconstruction using channel-dependent weights,

$$w_{\text{phys}} = \frac{N_{\text{expected}}}{N_{\text{generated}}}, \quad (4.4.1)$$

which account for the true interaction probability in the target for a given thickness L . This approach allows the use of oversampled histograms for shape and resolution studies while preserving the correct physical normalization of event yields.

4.5 Signal and background yields

For each target thickness L , signal and background yields are extracted using a counting approach in an adaptive invariant-mass window centered at the fitted peak position $\mu(L)$,

$$|m_{\gamma\gamma} - \mu(L)| < n_\sigma \sigma_m(L).$$

The number of reconstructed signal and background events in this window is computed independently from the signal-only and background histograms. The raw counts are subsequently rescaled using the physical weights w_{phys} defined in eq. 4.4.1 to obtain the expected signal and background yields, denoted as $S(L)$ and $B(L)$, respectively. This adaptive window ensures a consistent definition of signal and background across different target thicknesses, despite the variation of the mass resolution.

4.6 Uncertainty estimation via Poisson bootstrap

Statistical uncertainties on the extracted quantities are estimated using a Poisson bootstrap technique. For each invariant-mass histogram, bootstrap replicas are generated by fluctuating the content of each bin according to a Poisson distribution,

$$n_i^* \sim \text{Poisson}(n_i),$$

where n_i is the original bin content.

For each bootstrap replica, the full analysis chain is repeated, including the Gaussian fit, the adaptive counting procedure, and the computation of $S(L)$, $B(L)$ and derived quantities. The statistical uncertainty on each observable is then estimated as the standard deviation of its bootstrap distribution.

This method provides a robust estimate of uncertainties and naturally accounts for correlations between fit parameters, counting windows and reconstructed yields. The same bootstrap procedure is applied to estimate the error of the mass resolution $\sigma_m(L)$, ensuring a consistent treatment of uncertainties across all extracted observables.

5 Optimization of the target thickness

The target thickness L plays a central role in determining the experimental performance of the setup. On the one hand, increasing L raises the probability that an incoming π^- undergoes a hadronic interaction inside the material, thus enhancing the absolute production rate of π^0 and η mesons. On the other hand, a thicker target reduces the probability that both decay photons escape the material without being absorbed or merged, and degrades the mass resolution through increased path length and detector effects.

The optimization of L therefore results from a trade-off between interaction probability, photon escape probability, reconstruction efficiency, background contamination and invariant-mass resolution.

5.1 Signal and background reconstruction as a function of target thickness

For each target thickness L , oversampled signal samples for the $\pi^0 \rightarrow \gamma\gamma$ and $\eta \rightarrow \gamma\gamma$ channels and oversampled background sampled are generated and reconstructed as described in Section 4.

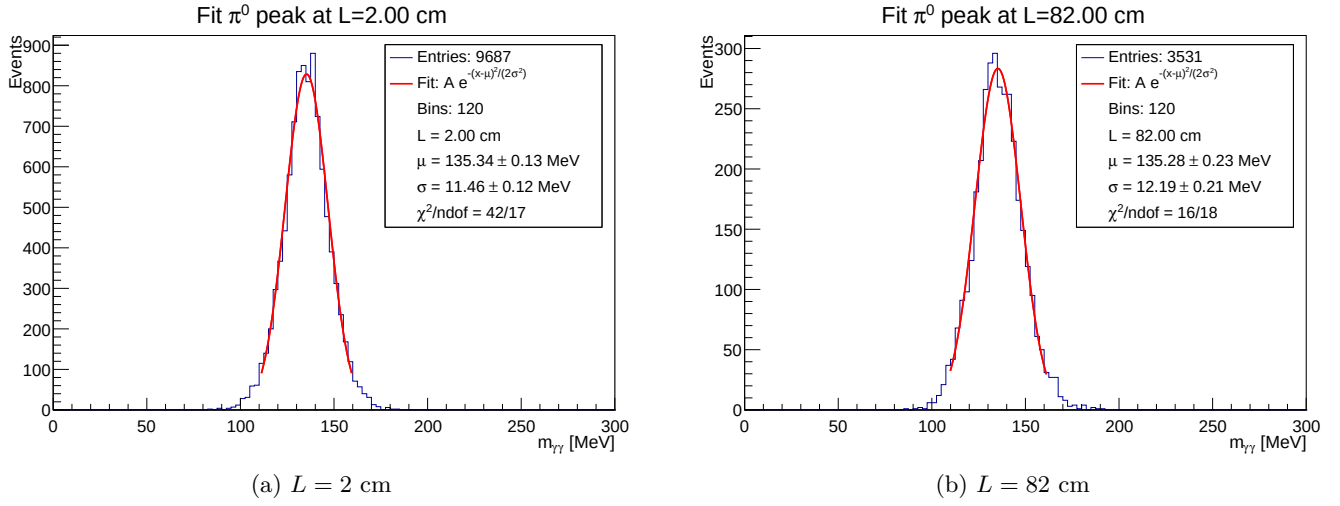


Figure 5.1.1: Reconstructed invariant-mass spectrum for the $\pi^0 \rightarrow \gamma\gamma$ channel at two representative target thicknesses. Total generated events: $N_{\text{generated}} = 10^4$.

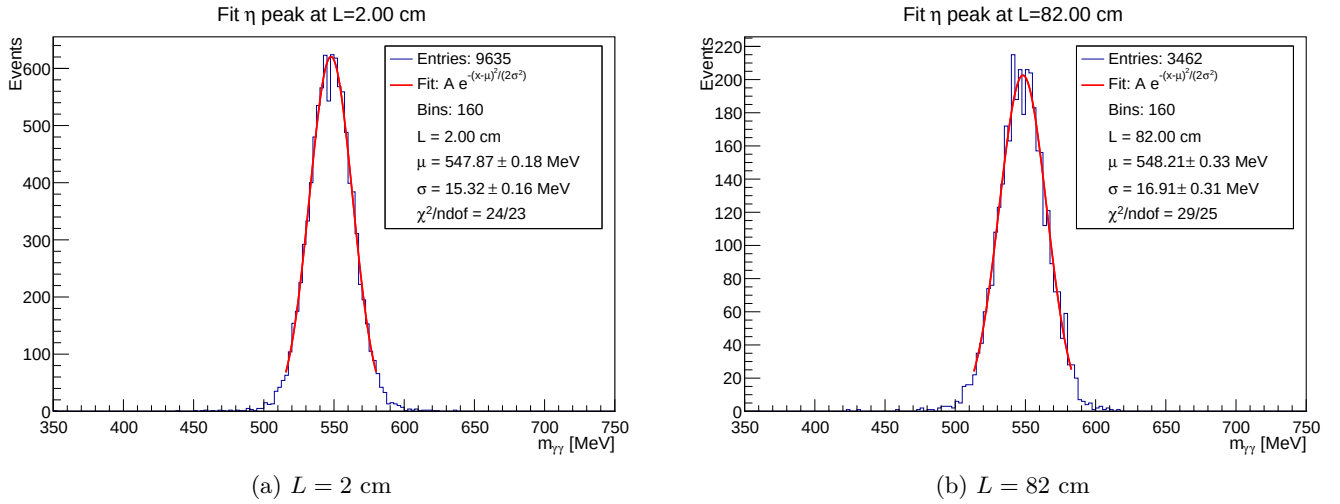


Figure 5.1.2: Reconstructed invariant-mass spectrum for the $\eta \rightarrow \gamma\gamma$ channel at two representative target thicknesses. Total generated events: $N_{\text{generated}} = 10^4$.

5.1.1 Signal reconstruction

Representative invariant-mass spectra and fits for small and large values of L for π^0 and η are shown respectively in Fig. 5.1.1 and Fig. 5.1.2 with a total number of oversampled events $N_{\text{generated}} = 10^4$. Two clear trends emerge:

- The total number of reconstructed signal events decreases with increasing L . This behavior is driven by the photon escape probability: photons produced deeper in the target have a higher chance of being absorbed or failing the cluster-separation and acceptance requirements.
- The invariant-mass resolution deteriorates as L increases. This is mainly due to the larger average distance traveled by the photons inside the material, which amplifies the impact of angular and energy smearing on the reconstructed opening angle.

These effects are visible for both mesons, with a more pronounced impact on the η channel due to its higher mass and typically smaller opening angles at fixed energy.

5.1.2 Background reconstruction

Fig. 5.1.3 shows the reconstructed oversampled background invariant-mass spectra with a total number of events $N_{\text{generated}} = 10^4$ for two representative target thicknesses. In both cases, the background exhibits a smooth and featureless shape over the full $m_{\gamma\gamma}$ range, with no resonant structure. This behavior is expected for a background

composed of uncorrelated photons, and confirms that the reconstruction and detector effects do not introduce spurious peaks in the signal regions.

The strong enhancement of events at low invariant mass can be understood from the background generation model. The photon energies are sampled from a log-uniform distribution, which corresponds to a probability density described in eq. 4.2.1 implying that soft photons are produced with much higher probability than hard ones. Since the invariant mass of two photons is given by eq. 4.1.1, events with small photon energies naturally populate the low-mass region, even for moderate opening angles. The observed background spectrum is therefore fully consistent with the expected kinematics of two soft, uncorrelated photons.

As the target thickness increases, the total number of the reconstructed background events decreases. This effect is driven by the reduced probability that both photons escape the target and satisfy the reconstruction criteria when produced deeper in the material.

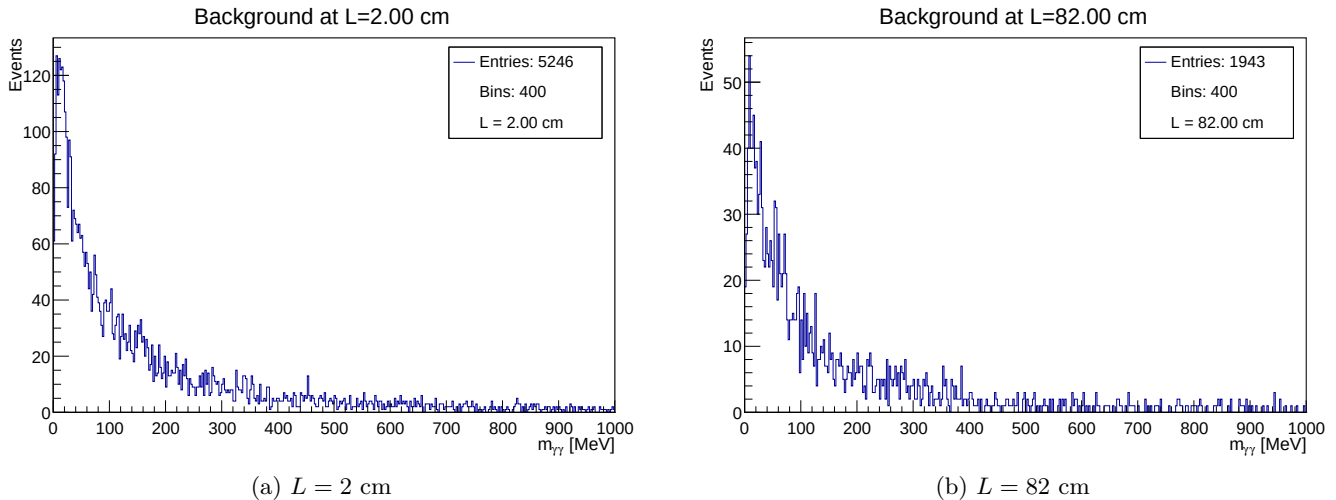


Figure 5.1.3: Reconstructed invariant-mass spectrum of the background for two representative target thicknesses. Total generated events: $N_{\text{generated}} = 10^4$.

5.2 Normalized spectra and signal-to-background comparison

To disentangle shape effects from absolute normalization, the reconstructed invariant-mass spectra for the π^0 , η , and background samples are normalized to unit area. This representation allows a direct comparison of the spectral shapes independently of production rates and reconstruction efficiencies.

Fig. 5.2.1 shows the normalized spectra for two representative target thicknesses. In both cases, the π^0 and η signals appear as narrow and well-defined peaks on top of a smooth background distribution. At the level of normalized spectra, the signal peaks are clearly visible and are not overwhelmed by the background.

Increasing the target thickness affects all reconstructed samples in a largely proportional way. Both signal and background yields decrease due to photon absorption and acceptance losses, while the signal peaks exhibit a modest broadening associated with the worsening mass resolution. As a consequence, the qualitative appearance of the normalized spectra remains similar for different values of L , and the difference in signal shapes between the two panels in Fig. 5.2.1 is not appreciable.

These plots demonstrate that the background dominance in the physically weighted spectra arises instead from the much larger total hadronic cross section compared to the exclusive π^0 and η production channels. This distinction highlights the complementary roles of detector resolution and production dynamics in the optimization problem: while the former governs the separability of signal and background in invariant mass, the latter ultimately controls the achievable statistical significance.

5.3 Definition of the significance

To quantify the sensitivity of the measurement as a function of the target thickness L , a simple counting-experiment significance is adopted. The significance is defined as

$$Z(L) = \frac{S(L)}{\sqrt{S(L) + B(L)}},$$

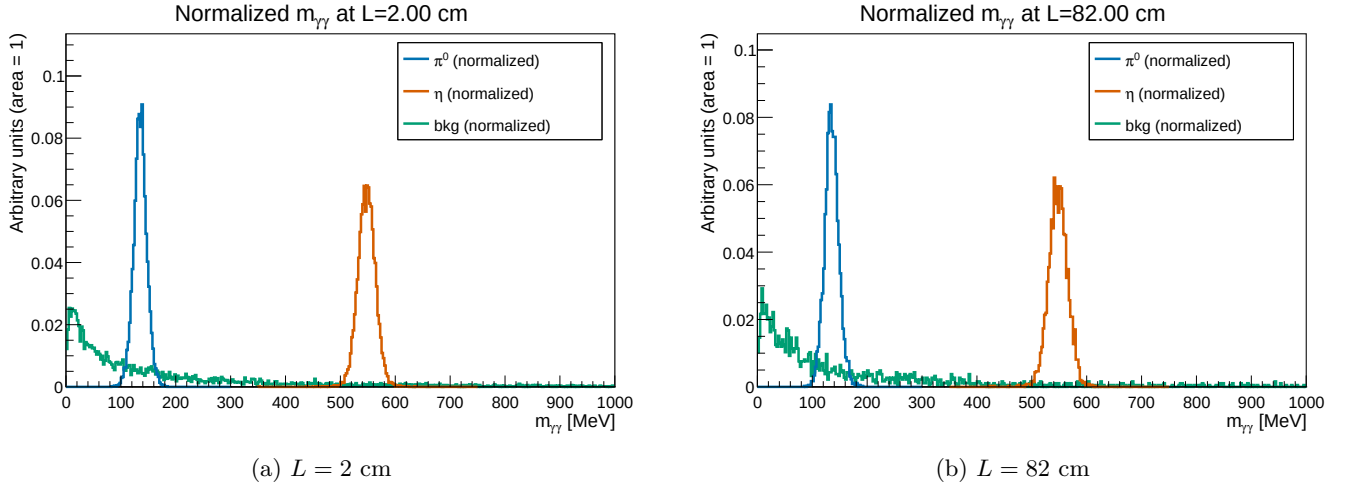


Figure 5.2.1: Invariant-mass spectra normalized to unit area for the π^0 , η , and background samples at two representative target thicknesses. The normalization removes the effect of absolute yields, allowing a direct comparison of spectral shapes.

where $S(L)$ and $B(L)$ denote the expected signal and background yields, respectively, within a fixed invariant-mass window around the reconstructed peak.

This expression can be interpreted in a simple counting-experiment picture. In a fixed invariant-mass window, the total number of observed events is $N = S + B$, which follows Poisson statistics. The corresponding statistical uncertainty is therefore $\sigma_N = \sqrt{S + B}$. The ratio $S/\sqrt{S + B}$ thus measures how large the signal yield is compared to the typical statistical fluctuations of the total event count. For this reason, it provides a direct estimate of the statistical distinguishability of the signal from the background.

The quantities $S(L)$ and $B(L)$ are obtained by reweighting the oversampled Monte Carlo histograms to the expected number of physical interactions in the target with error from Poisson bootstrap as described in Sections 4.4-4.6.

The resulting $Z(L)$ therefore incorporates the combined effects of interaction probability, photon escape efficiency, detector resolution, and background contamination.

5.4 Optimization results

The results of the target thickness scan are summarized in Figs. 5.4.1 and 5.4.2, which show the behavior of the significance, mass resolution, and reconstruction efficiency as a function of L for both the π^0 and η channels.

The significance remains small over the full range of target thicknesses for both mesons. This is a direct consequence of the strongly background-dominated regime, with $B(L) \gg S(L)$ at all values of L . As discussed in the previous sections, this behavior reflects the large difference between the total hadronic cross section and the much smaller exclusive π^0 and η production cross sections, rather than a limitation of the reconstruction or detector model.

Despite the low absolute values of $Z(L)$, the scan remains meaningful as a relative optimization. For small values of L , the interaction probability is low, resulting in a reduced signal yield. As L increases, the signal yield initially grows, leading to a moderate increase in the significance. At larger thicknesses, however, the gain in yield is offset by the simultaneous growth of the background and by the degradation of detector performance, causing the significance to saturate.

The mass resolution shows a clear worsening with increasing target thickness for both channels, as illustrated in Fig. 5.4.2. This effect is driven by the increasing average photon path length inside the target, which enhances the impact of energy and angular smearing on the reconstructed invariant mass. At the same time, the reconstruction efficiency decreases steadily with L due to photon absorption, acceptance losses, and cluster overlap effects. The background efficiency follows a similar trend, indicating that these losses are largely governed by photon transport rather than channel-specific kinematics.

6 Conclusions

The optimization of the target thickness highlights the existence of an intermediate range of L values for which the overall experimental performance is maximized. In this region, the increase in signal yield due to a higher

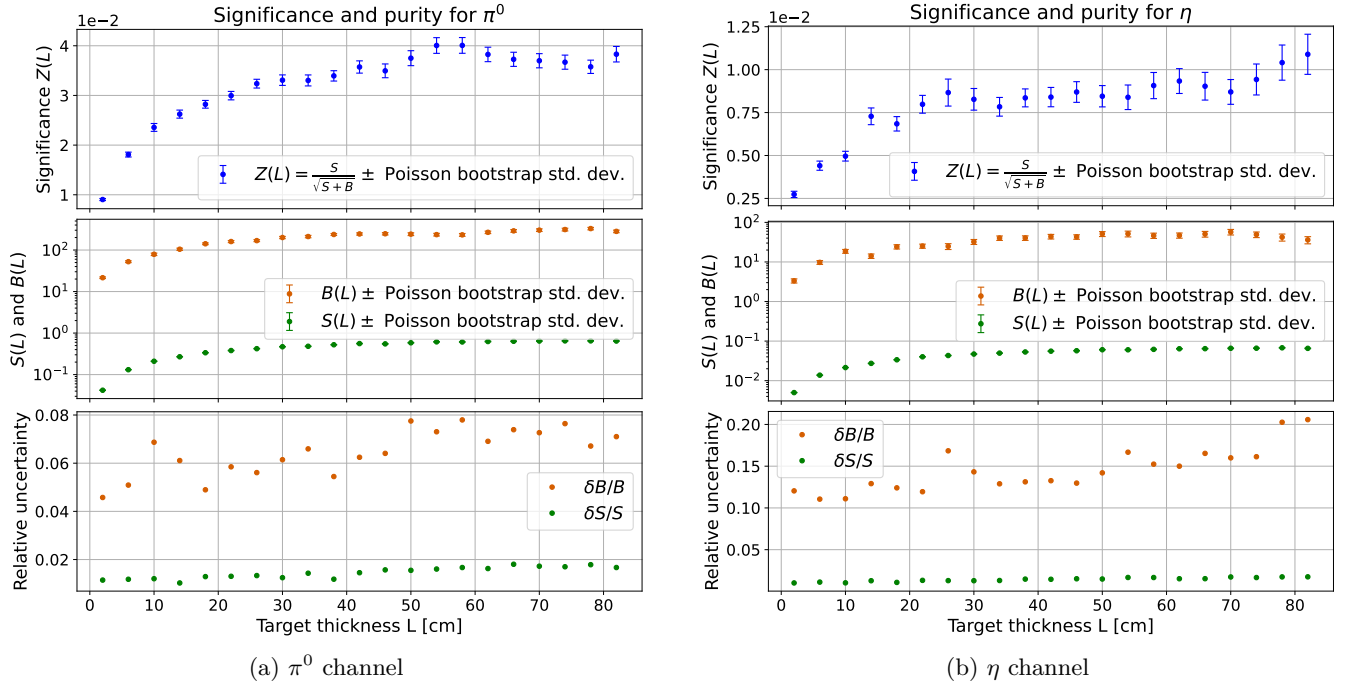


Figure 5.4.1: Significance $Z(L) = S/\sqrt{S+B}$ as a function of the target thickness L for the π^0 and η channels. The lower panels show the corresponding signal and background yields, together with their relative statistical uncertainties.

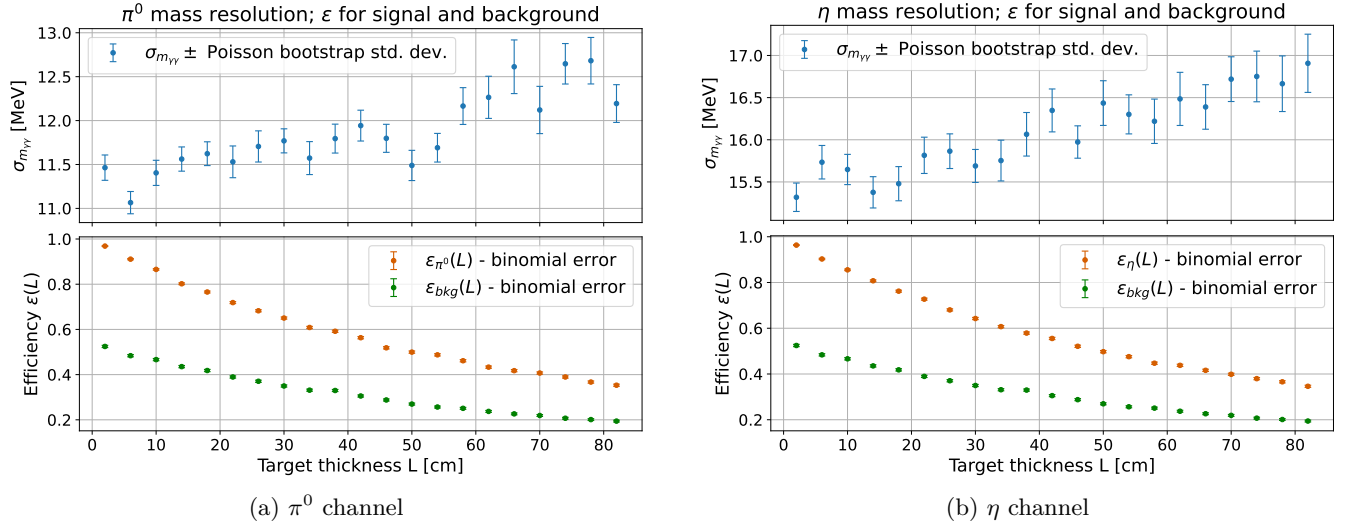


Figure 5.4.2: Invariant-mass resolution $\sigma_{m_{\gamma\gamma}}$ and reconstruction efficiency as a function of the target thickness L for the π^0 and η channels.

interaction probability is not yet outweighed by the combined effects of background growth, photon absorption, and degradation of the invariant-mass resolution.

Although the absolute significance remains small for all values of L , reflecting the intrinsically background-dominated nature of the process, the scan shows that, within the assumptions of the present detector model and cross-section inputs, an optimal target thickness lies in the range

$$L \simeq [40, 60] \text{ cm},$$

where the balance between signal yield, background contamination, and detector performance on the reconstructed mass resolution is most favorable.

This result emphasizes that the achievable sensitivity is ultimately limited by the underlying production dynamics and detector response, and that target optimization can improve, but not overcome, the constraints imposed by the relative sizes of signal and background cross sections and detector resolution.

A Validation of the interaction-depth sampling

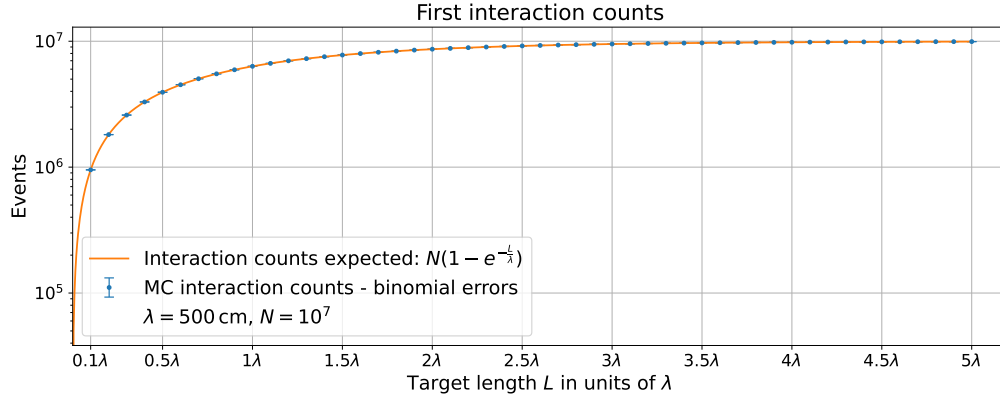


Figure A.0.1: Comparison between the Monte Carlo sampled number of first interactions and the analytically expected behaviour as a function of the target thickness L . Error bars reflect binomial statistical uncertainties.

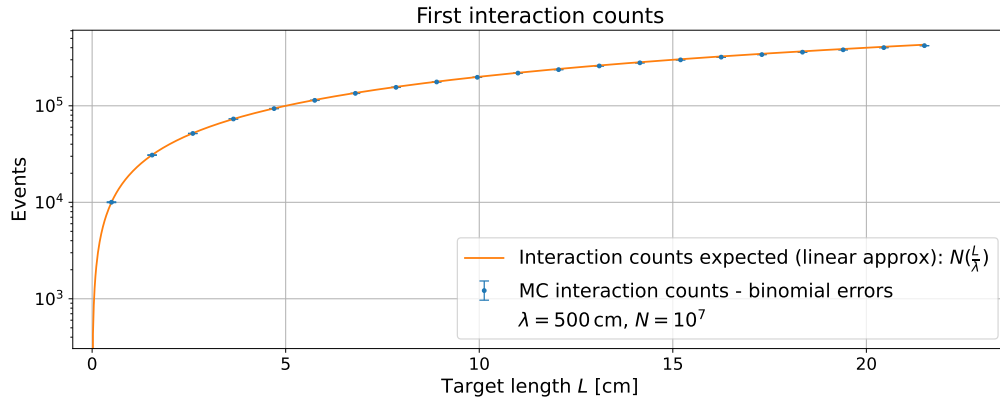


Figure A.0.2: Comparison between the Monte Carlo sampled number of first interactions and the analytically expected behaviour as a function of the target thickness L in the case of $L \ll \lambda$. Error bars reflect binomial statistical uncertainties.

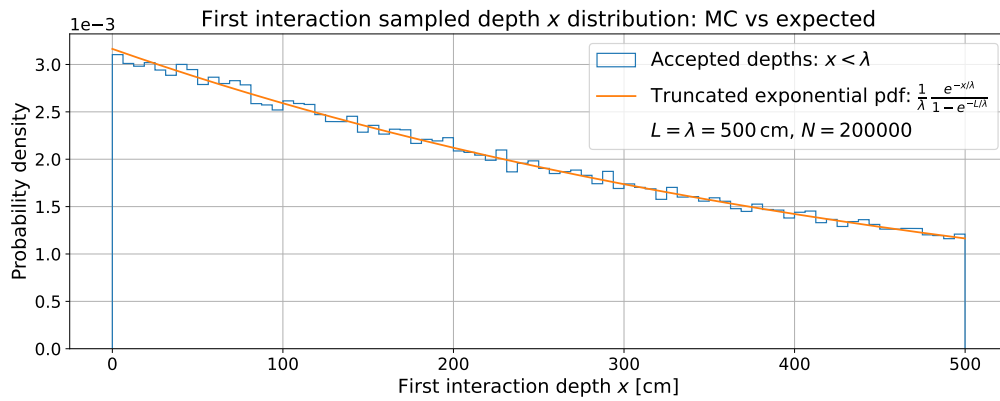


Figure A.0.3: Distribution of the sampled interaction depth z for a fixed target thickness, compared to the corresponding truncated exponential probability density. Small deviations are due to finite Monte Carlo statistics and binning effects.

B Poisson confidence intervals and Garwood construction

In counting experiments, the number of observed events n is commonly modeled as a Poisson-distributed random variable with mean μ ,

$$P(N = n \mid \mu) = \frac{\mu^n e^{-\mu}}{n!}.$$

The standard estimate $\hat{\mu} = n$ corresponds to the maximum-likelihood estimator of the Poisson mean, with associated error $\sqrt{n}$ from the variance point estimation.

This estimation becomes unreliable in the low-count regime, because the relative fluctuations of the estimator are large and the distribution of possible outcomes is strongly asymmetric.

For this reason, a more appropriate statistical description is provided by interval estimation based directly on the Poisson likelihood.

A confidence interval $[\mu_{\text{low}}, \mu_{\text{up}}]$ at confidence level $1 - \alpha$ can be defined by inverting the cumulative Poisson distribution. For an observed count n , the interval boundaries are obtained by solving

$$P(N \leq n \mid \mu_{\text{up}}) = 1 - \frac{\alpha}{2}, \quad P(N \geq n \mid \mu_{\text{low}}) = 1 - \frac{\alpha}{2}.$$

The Poisson cumulative distribution can be expressed in terms of the incomplete gamma functions,

$$P(N \leq n \mid \mu) = \frac{\gamma(n+1, \mu)}{\Gamma(n+1)}, \quad P(N \geq n \mid \mu) = \frac{\Gamma(n, \mu)}{\Gamma(n)},$$

where γ and Γ denote the lower and upper incomplete gamma functions, respectively.

These gamma functions are directly related to the cumulative distribution of the χ^2 variable. In particular, the identity

$$\frac{\gamma(k, \mu)}{\Gamma(k)} = F_{\chi^2}(2\mu; 2k)$$

allows the Poisson confidence interval to be written in closed form using χ^2 quantiles.

As a result, the Garwood confidence interval for the Poisson mean μ is given by

$$\mu_{\text{low}} = \frac{1}{2} \chi_{\alpha/2}^2(2n), \quad \mu_{\text{up}} = \frac{1}{2} \chi_{1-\alpha/2}^2(2(n+1)),$$

where $\chi_p^2(\nu)$ denotes the p -quantile of the χ^2 distribution with ν degrees of freedom.

For the case $n = 0$, the lower confidence bound is not defined. Indeed, the condition

$$P(N \geq 0 \mid \mu) = 1$$

is satisfied for any value of μ , implying that no finite lower bound can be set. Consequently,

$$\mu_{\text{low}}(n = 0) = 0,$$

while the upper bound remains well defined,

$$\mu_{\text{up}}(n = 0) = \frac{1}{2} \chi_{1-\alpha/2}^2(2).$$

This asymmetric behaviour reflects the physical fact that observing zero events constrains the Poisson mean from above, but does not exclude arbitrarily small values.

C Log-uniform energy sampling

Let E be a continuous random variable constrained to the interval $[E_{\text{min}}, E_{\text{max}}]$ with $0 < E_{\text{min}} < E_{\text{max}}$. It can be said that E is log-uniform on $[E_{\text{min}}, E_{\text{max}}]$ if the logarithm of the energy,

$$Y = \log E,$$

is uniformly distributed:

$$Y \sim U(\log E_{\text{min}}, \log E_{\text{max}}).$$

Equivalently, the probability density of Y is constant on its support,

$$p_Y(y) = \begin{cases} \frac{1}{\log(E_{\max}/E_{\min})}, & y \in [\log E_{\min}, \log E_{\max}], \\ 0, & \text{otherwise.} \end{cases}$$

The PDF of E can be obtained from the standard change-of-variables formula. Since $y = \log E$ and $dy/dE = 1/E$, one has

$$p_E(E) = p_Y(\log E) \left| \frac{d(\log E)}{dE} \right| = p_Y(\log E) \frac{1}{E}.$$

Substituting the constant p_Y yields

$$p_E(E) = \begin{cases} \frac{1}{E \log(E_{\max}/E_{\min})}, & E \in [E_{\min}, E_{\max}], \\ 0, & \text{otherwise.} \end{cases}$$

Therefore, the log-uniform distribution corresponds to a scale-free spectrum $p_E(E) \propto 1/E$.

A key property of the log-uniform distribution is that it assigns equal probability to equal intervals in $\log E$. In particular, consider an interval spanning one decade in base-10, i.e. $[E_0, 10E_0]$. The probability of falling in that interval is

$$\mathbb{P}(E \in [E_0, 10E_0]) = \int_{E_0}^{10E_0} p_E(E) dE = \frac{1}{\log(E_{\max}/E_{\min})} \int_{E_0}^{10E_0} \frac{dE}{E} = \frac{\log 10}{\log(E_{\max}/E_{\min})}.$$

This result is independent of E_0 , hence each energy decade carries the same probability mass (as long as the decade lies within $[E_{\min}, E_{\max}]$). In Monte Carlo terms, for a fixed total number of generated photons, each decade contains on average the same number of events.

The cumulative distribution function (CDF) is

$$F(E) \equiv \mathbb{P}(E' \leq E) = \int_{E_{\min}}^E p_E(E') dE' = \frac{1}{\log(E_{\max}/E_{\min})} \int_{E_{\min}}^E \frac{dE'}{E'} = \frac{\log(E/E_{\min})}{\log(E_{\max}/E_{\min})}, \quad E \in [E_{\min}, E_{\max}].$$

To generate E according to p_E , one uses the inverse transform method: if $u \sim U(0, 1)$ is a uniform random number, set $u = F(E)$ and solve for E :

$$u = \frac{\log(E/E_{\min})}{\log(E_{\max}/E_{\min})} \implies \log\left(\frac{E}{E_{\min}}\right) = u \log\left(\frac{E_{\max}}{E_{\min}}\right) \implies E = E_{\min} \exp\left[u \log\left(\frac{E_{\max}}{E_{\min}}\right)\right].$$

This can be rewritten in the numerically convenient form

$$E = E_{\min} \left(\frac{E_{\max}}{E_{\min}} \right)^u.$$

This expression is used in the toy background generator to sample photon energies exactly from the log-uniform distribution.

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